

Competition and Pass-Through: Evidence from Isolated Markets[†]

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We measure how pass-through varies with competition in isolated oligopolistic markets with captive consumers. Using daily pricing data from gas stations on small Greek islands, we study how unanticipated and exogenous changes in excise duties are passed through to consumers in markets with different numbers of retailers. We find that pass-through increases from 0.4 in monopoly markets to 1 in markets with four or more competitors and remains constant thereafter. The speed of price adjustment is about 60 percent higher in more competitive markets. Finally, we show that geographic market definitions based on measures of distance across sellers often result in significant overestimation of the pass-through. (JEL H22, H25, L13, L81)

A fundamental issue in economics is how firms pass cost shocks (taxes, exchange rates, input prices) through to prices. The incidence and pass-through of taxation are classic public policy concerns (Fullerton and Metcalf 2002). The pass-through of exchange rates and tariffs has important repercussions on firm productivity and international trade (De Loecker and Koujianou-Goldberg 2014). Moreover, the pass-through of input prices is relevant to the analysis of oligopolistic markets, price discrimination (Aguirre, Cowan, and Vickers 2010), and merger analysis (Jaffe and Weil 2013). Finally, the cost pass-through is also relevant to the policy debate (Genakos, Grey, and Ritz 2020) in many industries, such as the health (Cabral, Geruso, and Mahoney 2015) and energy sectors (Fabra and Reguant 2014).

Theoretical analysis shows that competition is a key determinant of pass-through (Weyl and Fabinger 2013). As for empirical analysis, there is a well-established line of research exploiting plausibly exogenous variability in costs to infer the magnitude

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of pass-through.¹ However, there is little evidence on the relation between competition and pass-through. In empirical studies, competition is generally measured by the number of competitors located within a given geographic area around each firm. Variability in the number of close competitors captures some important aspects of competition but does not guarantee that there are no significant substitution effects beyond the selected geographic area. For example, some consumers may commute across geographic markets for work or family reasons, hence generating substitution effects across geographically distant markets. Although market structure is recognized as an endogenous outcome, it is generally difficult to find valid instruments for competition.

We measure how pass-through varies with competition in isolated oligopolistic markets of different sizes. Our data come from the retail market for petroleum products (gasoline, unleaded gasoline, diesel, heating oil) on the many small islands of Greece. Some of these islands are so small that they have only a single gas station, while others have two, three, or more. The naturally occurring variability in island size provides an exogenous source of variability in the level of competition. Islands clearly define local markets, as substitution effects between islands are zero.² The logic of this approach is similar to that of Bresnahan and Reiss (1991), who study entry in geographically isolated oligopolistic markets.

Along with this unique setting, we take advantage of significant policy changes made by the Greek government when, at the beginning of the financial crisis in 2010, they increased the excise duty on petroleum products three times. The increments were large and unannounced and provide us with an ideal exogenous shock for estimating the pass-through to retail prices. For political reasons, heating oil was excluded from the excise hikes, as it was considered a necessity good.

Using daily gas station data, we study how the pass-through of the excise duty varied across markets with different numbers of competitors while using heating oil as a control group. Thus we can account for unobserved heterogeneity across islands and gas stations and control for the daily aggregate price fluctuations of petroleum products using the control group. We find four main results. First, the response to cost shocks is rapid, with an average pass-through of 0.71 after ten days from an excise duty change. Second, pass-through increases significantly with the number of competitors, and the relation between competition and pass-through is nonlinear. On average, the pass-through is 0.43 on monopolistic islands and grows to about 1 on islands with four or more competitors. Third, we find that price adjustments are larger and occur more quickly in more competitive markets, leading to faster pass-through in more competitive markets. Fourth, we find that using geographic market definitions based on distance across sellers (rather than the island market definition) results in overestimation of pass-through in highly concentrated markets.

¹ Such variability may come from changes in sales taxes (Barzel 1976; Poterba 1996; Besley and Rosen 1998; Marion and Muehlegger 2011), exchange rate fluctuations (Campa and Goldberg 2005; Gopinath, Gourinchas, Hsieh, and Li 2011), or changes in input prices (Borenstein, Cameron, and Gilbert 1997; Genesove and Mullin 1998; Nakamura and Zerom 2010; Miller, Osborne, and Sheu, 2017).

² Refueling a car by traveling to a different island is prohibitively expensive, and privately importing fuel in tanks or similar containers is dangerous and illegal.

Our results contribute to the literature on the transmission of cost shocks to prices whose ultimate objective is to understand the strength of nominal rigidities and the impact of fiscal, monetary, and exchange rate policy. Existing evidence on the impact of competition on pass-through is scarce and somewhat mixed. For example, Miller, Osborne, and Sheu (2016) find that increasing competition reduces pass-through in a market in which the pass-through is above unity.³ Cabral, Geruso, and Mahoney (2018) study the pass-through of government subsidies to premiums of Medicare Advantage plans and find evidence of larger pass-through in more competitive markets, with pass-through estimates ranging between 13 percent and 74 percent. Alm, Sennoga, and Skidmore (2009) find a somewhat lower pass-through in rural than in urban gasoline markets, which might be related to the different competitive environments. In contrast, Doyle and Samphantharak (2008) and Stolper (2018) find that greater brand concentration and market power are associated with larger pass-through rates in the gasoline market.

Our finding of a quick response to cost shocks is in line with the results of Bonadio, Fisher, and Sauré (2020), who show a two-week adjustment period after a large exchange rate shock. Our results imply a positive correlation between pass-through and frequency of price adjustments across markets with different levels of competition, which is in line with the results of Gopinath and Itskhoki (2010). Our results are also related to the literature on the nonlinear effects of competition on firm behavior (Bresnahan and Reiss 1991) by showing that the pass-through quickly converges to competitive levels as the number of competitors grows. Finally, the paper contributes to the empirical literature on the estimation of cost pass-through and the analysis of competition in geographic markets (Houde 2012).

I. Theoretical Background

Economic theory provides some general results on how competition and other variables interact in determining the level of pass-through. Following the conduct parameter approach of Genesove and Mullin (1998), Weyl and Fabinger (2013) obtain an equation for the pass-through in oligopolistic markets with n symmetrically differentiated firms. Denoting the elasticity of demand by $\epsilon_D = -p/qp'$, they describe the solution to the firm maximization problem by a conduct parameter $(p - mc(q))/p\epsilon_D = \theta$, where $mc(q)$ is the marginal cost. θ captures the intensity of the competition among firms ($\theta = 0$ in a competitive market and $\theta = 1$ in a monopolistic market). Independently of the specific model considered, the impact of an increase in marginal cost (i.e., the pass-through) on the equilibrium price is

$$(1) \quad \rho = \frac{1}{1 + \frac{\theta}{\epsilon_\theta} + \frac{\epsilon_D - \theta}{\epsilon_S} + \frac{\theta}{\epsilon_{ms}}}.$$

³This is consistent with convergence toward a unit pass-through as competition increases. Other studies have found evidence of pass-through larger than one (see, for example, Besley and Rosen 1999; Hanson and Sullivan 2009; Bonnet and Réquillart 2013), which is entirely possible according to theory (see Section I).

The pass-through ρ depends on the conduct parameter θ and how it varies as the quantity produced changes ($\epsilon_\theta = (\theta/(q(d\theta/dq)))$) but also on the determinants of the elasticity of demand ϵ_D , the elasticity of the inverse marginal cost curve ϵ_S (the elasticity of supply), and the curvature of the demand function ϵ_{ms} .⁴ In general, the sign and magnitude of the pass-through is ambiguous. The sign of the effect of an increase in the conduct parameter on the pass-through can be either positive or negative.

The expression for ρ greatly simplifies in some special cases, highlighting the role of the different elements in the denominator of equation (1). The ratio $(\epsilon_D - \theta)/\epsilon_S$ links demand and supply elasticity and pass-through.⁵ This ratio is equal to zero if the marginal cost is constant. As we will argue in Section II, it is realistic to assume that marginal cost is constant at the firm level, at least in the short run and for the range of quantities typically sold by gas stations in our sample. This suggests that demand heterogeneity is unlikely to play a big role in our application through the ratio $(\epsilon_D - \theta)/\epsilon_S$. Still, heterogeneity in demand curvature may be relevant through the ratio θ/ϵ_{ms} .

A second interesting special case is when θ is constant. If θ is constant, then the term θ/ϵ_θ is also equal to zero. The conduct parameter θ is a constant in a number of prominent models. For example, θ is equal to 1 in monopoly, equal to 0 in perfect competition and in the Bertrand model, and equal to $1/n$ in the Cournot model.⁶ The conduct parameter is assumed to be a constant in most empirical applications based on the conduct parameter approach.

Finally, an important determinant of the pass-through is the demand curvature ϵ_{ms} . Many empirical studies are based on linear demand specifications, directly implying that $\epsilon_{ms} = 1$. However, it is not uncommon to assume different demand specifications that imply different curvature, although there is little guidance in the literature on the sign and magnitude of ϵ_{ms} .

If the marginal cost were constant, θ were constant, and demand were linear, then $\rho = 1/(1 + \theta)$ and an increase in the conduct parameter (less competition) would lead to lower pass-through.⁷ The first assumption is met in several industries and is realistic in our application. The second is often considered a reasonable simplification in empirical studies in not putting restrictions on the intensity of competition. However, the third is difficult to justify without specific evidence on the second derivative of the demand function. Hence, in general, the impact of an increase in competition on pass-through remains largely an empirical issue.

⁴ $\epsilon_{ms} = \frac{ms}{ms'q}$, where ms is the negative of the marginal consumer surplus ($ms = -p'q$). ϵ_{ms} measures the curvature of the log of demand (Fabinger and Weyl 2012). If demand is linear, then $\epsilon_{ms} = 1$; if concave, $\epsilon_{ms} < 1$; and if convex, $\epsilon_{ms} > 1$ (and the opposite is also true).

⁵ Note that if $\theta = 0$, then $\rho = \frac{1}{1 + (\epsilon_D/\epsilon_S)}$, which is the classic formula for tax pass-through in perfect competition.

⁶ The relation between the conduct parameter and the number of firms n illustrates the sense in which an increase in the number of firms leads to greater competition. In empirical papers, which typically deal with specific industries, the number of firms is often used as a proxy for the intensity of competition.

⁷ Moreover, in this special case the relation between pass-through and θ can be inverted, so estimating the pass-through provides information about the conduct parameter.

II. Industry Background and Data

Oil is the main energy source in Greece. In 2010 it accounted for 52 percent of the country's total primary energy supply, which is substantially higher than the average in most other advanced countries (36 percent in 2010).⁸ Two companies operate in the Greek refining industry: Hellenic Petroleum has three refineries while Motor Oil Hellas has one. Hellenic Petroleum controls 72 percent of the wholesale market.⁹ There are ten oil terminals in Greece, seven of which are in the Attica area (Athens) and three in the Salonica area (north). In 2010 there were 20 fuel trade companies operating in the retail market, the largest of which were EKO (a subsidiary of Hellenic Petroleum), Shell, BP, Avin Oil (100 percent subsidiary of Motor Oil), and Jet Oil. The marginal cost of petroleum products depends on long-term contracts between gas stations and trade companies. Within the observed range of quantities sold, the marginal cost of gas stations is reasonably constant. EU member states are required to impose a minimum array of energy taxes, but each member state has significant freedom in setting tax rates.¹⁰ There are two main taxes that are imposed on energy products: excise duties, which are a unit tax rate (euro cents per liter), and the Value Added Tax (VAT), which is a percentage tax. The retail price is determined as $P_{\text{retail}} = (P_{\text{refinery}} + \text{taxes \& fees} + \text{margins})(1 + \text{VAT})$. In this paper, we focus on the impact of changes in excise duties on prices, which are reported net of VAT.

In 2010 the inability of the Greek government to borrow new funds from international markets led to financial support from euro-area member states and the International Monetary Fund. One of the first measures taken by the Greek government to increase tax revenues was to increase excise duties on fuel. Excise duties on fuel were raised three times in 2010. Each of these three tax changes were announced and implemented the day after the decision was made as typically happens in order to reduce opportunities for arbitrage. Table 1 shows that the tax changes were significant (between 8 percent and 29 percent) and different across products. Remarkably, excise duties for heating oil remained unchanged.¹¹

Data and Measurement of Competition.—We combined several datasets for our analysis. First, we obtained daily station-level retail prices during 2010 for a sample of gas stations located on Greek islands. The data contain information on five different gasoline products: unleaded 95, unleaded 100, super (or leaded gasoline), diesel, and heating oil. The data on prices were officially collected by the Greek Ministry of Development and Competitiveness (2018) through a reporting system, which required managers of each petrol station to record retail prices

⁸International Energy Agency, Energy Policies of IEA Countries, 2011 review.

⁹The Greek government owns 35.5 percent of Hellenic Petroleum but no shares in Motor Oil Hellas.

¹⁰EU guideline 2003/96/EU.

¹¹Heating oil is chemically identical to diesel (although colored differently to prevent substitution) and is sold by the same gasoline stations throughout the country. A lower excise duty is applied to it as it is considered a necessity, since the vast majority of households use heating oil rather than gas during the winter months. It is illegal to sell and use heating oil for transportation and the prohibition is actively enforced.

TABLE 1—EXCISE DUTY CHANGES (EURO CENTS PER LITER AND $\Delta\%$)

Type of energy product	Unleaded 95 (1)	Unleaded 100 (2)	Diesel (3)	Super (leaded) (4)	Heating oil (5)
Before	41	41	30.2	42.1	2.1
February 10, 2010	53 (29%)	53 (29%)	35.2 (17%)	54.1 (29%)	2.1 (0%)
March 4, 2010	61 (15%)	61 (15%)	38.2 (9%)	62.1 (15%)	2.1 (0%)
May 3, 2010	67 (10%)	67 (10%)	41.2 (8%)	68.1 (10%)	2.1 (0%)

Notes: The table reports the level and percentage changes in excise duties by product.

Source: Authors' calculations based on data from the Eurostat (rates and structure of excise duties for energy products).

daily. The purpose of this system was to facilitate comparison and reduce search costs for consumers.

Second, we obtained socioeconomic (e.g., education, income, number of tourist arrivals) and geographic (size, distance from Piraeus,¹² distance from mainland, number of ports and airports, etc.) characteristics of each island from the Hellenic Statistical Authority (2010). Third, we measured the number of gas stations operating in each island using independent information from Yellow Pages (2018) and company reports.¹³ Finally, using Google Maps (2018), we geolocated each gas station and verified whether each station was offering any additional services (such as a shop, car wash, tire repairs, etc).

We focus on the 33 smallest islands, with fewer than 8 stations (Table A1 in the online Appendix), and a period of 10 days before and after each excise duty change described in Table 1. The price-reporting system provides more than 10,000 observations at the product–station level for 58 gas stations (or 61 percent of all stations) located on 26 islands. We test the representativeness of the final sample of gas stations and cannot reject the null hypothesis of no difference in mean characteristics between stations in and out of the sample.¹⁴ Table 2 reports summary statistics.

There are two main reasons for focusing on small islands. First, small islands have a small population and are physically small. The median island in our sample has about 2,500 inhabitants, and it is just 74 square kilometers (Table 2). Hence, consumers plausibly have close to perfect information about each station's prices and can reach all of them quite easily. Second, the number of competitors on a given island is the result of an entry game. In equilibrium, larger islands can sustain more competitors, each of them enjoying smaller markups. Bresnahan and Reiss (1991) find that competitive conduct changes quickly as the number of incumbents

¹² Piraeus is the primary distribution center for gasoline products in Greece.

¹³ Industry reports and Yellow Pages data for different years show that entry and exit was essentially zero during this period.

¹⁴ We compute the average number of competitors on the island, average island population, size (in square kilometers), number of tourist arrivals for that year, educational attainment of the island population, number of ports, number of airports, and station characteristics such as the presence of a shop, a car wash, or tire repair facilities for both groups. The results are reported in Table A2 in the online Appendix.

TABLE 2—SUMMARY STATISTICS

Variable	Mean	SD	Median	10th percentile	90th percentile
<i>Panel A. Prices (N = 10,129)</i>					
Unleaded 95 (euro cents per liter)	126	12.4	125	107	142
Unleaded 100 (euro cents per liter)	136	12.7	136	119	152
Super (euro cents per liter)	127	12.5	125	110	143
Diesel (euro cents per liter)	107	8.5	106	96	118
Heating oil (euro cents per liter)	62	4.1	62	57	67
<i>Panel B. Island characteristics (N = 33)</i>					
Size (km ²)	91	72	74	25	195
Population (number of inhabitants)	3,375	3,009	2,590	765	7,917
Ports	2	1	1	1	3
Airports	0.2	0.4	0.0	0.0	1.0
Arrivals (number of tourists)	120,069	173,642	60,785	15,411	296,016
Distance from Piraeus (km)	117	59	105	45	205
Income (€)	17,881	1,874	17,257	15,604	20,471
Education (tertiary, % population)	11%	2%	10%	9%	13%

Notes: Island socioeconomic and geographic characteristics were obtained from the Hellenic Statistical Authority (Census 2010). Arrivals refer to tourist arrivals by air or sea in 2010. Income per capita is based on a release from the Independent Authority of Public Revenue.

Source: Authors' calculations based on data from the Greek Ministry of Development and the Hellenic Statistical Authority.

increases. They find that the most variation in conduct occurs with the entry of the second or third firm, hence selecting islands with fewer than eight firms provides a sufficiently large range to capture the main effects of competition.

Arbitrage across islands is basically impossible, as the cost of transporting a car by ferry outweighs the potential savings in fuel costs. The set of islands considered in this paper generally host only small fishing boats, which are too slow to make it profitable to travel to a different island for refueling. Larger vessels (commercial ships or larger fishing boats) stop and refuel in larger ports with more facilities and possibly connections with the mainland.

Different measures of competition are possible for islands with more than one gas station. Having geolocated each station, we also compute measures of competition based on the number of competitors within a three-kilometer driving distance from each station, a three-kilometer radius, or a ten-minute driving time, thus taking into consideration not just the distance but also the underlying geography of the island. These are conventional methods of measuring competition when there is no natural boundary across markets. In Section IVD we compare the results obtained using these alternative measures. Finally, there is very little to no brand concentration at the island level. In all islands in our sample, gas stations are either franchisees of different brands (dealer operated, not company owned) or independently owned.¹⁵ Hence, the number of gas stations on each island realistically captures the number of competitors on that island.

¹⁵ Our results will not be affected by the type of gas station or the type of contract.

Islands vary in size and the number of gas stations. Figure 1, panel A shows that the larger the island (in terms of area or population), the larger the number of stations. Monopolies have, on average, about 1,100 inhabitants while islands with seven stations have about 9,800 inhabitants. In terms of physical size, monopolies are 54 square kilometers on average while islands with seven stations are about 110 square kilometers. Prices vary significantly across islands. For example, Figure 1, panel B shows the distribution of the average prices for diesel and heating oil across islands.¹⁶ Prices tend to fall, on average, as the number of competitors increases. Finally, Figure 1, panel C shows the negative correlation between island size and prices. Taken together, Figure 1 shows that larger islands tend to support more competitive markets, thus leading to lower prices. Gas stations on small islands tend to change prices less frequently than on the mainland. In our sample a station adjusted the price of a product every five days on average throughout 2010.

III. Identification and Empirical Methodology

We use a difference-in-difference approach, and we start by estimating the following model:

$$(2) \quad P_{kist} = \beta_0 + \rho Tax_{kt} + \beta_{ks} + \beta_t + e_{kist},$$

where P_{kist} denotes the retail price of product k , on island i , in gas station s , on day $t \in \{\tau - 1, \tau + \delta\}$, where τ is the date of each of the three excise duty changes and $\delta = 1, \dots, 10$ is the length of the adjustment period considered. Tax_{kt} is the excise duty, and the coefficient ρ captures the tax pass-through. Finally, the model includes product–gas station and day fixed effects. This econometric approach follows a long literature on difference-in-difference estimators and is based on the comparison of prices on two different dates (before and after the policy change) for a treatment group (gasoline and diesel) and a control group (heating oil).¹⁷

We then focus on the interaction between taxes and competition and estimate the model

$$(3) \quad P_{kist} = \beta_0 + \rho(n_i, Z_i) Tax_{kt} + \beta_{ks} + \beta_t + e_{kist},$$

where the pass-through $\rho(n_i, Z_i)$ is a linear function $\rho(n_i, Z_i) = \rho_0 + \rho_1 n_i + \rho_2 Z_i$ of the number of competitors n_i and other island-specific characteristics Z_i . Alternatively, the relation between pass-through ρ and number of stations j can be nonparametrically estimated replacing $\rho(n_i) = \sum_j \rho_j I(n_i = j)$, where I is an indicator variable for each observed number of gas stations.

The identifying assumption is $E(e_{kist}|X) = 0$, where X is the matrix of all covariates. This OLS condition is reasonably met in our difference-in-difference

¹⁶ The range of prices in Figure 3 is about €0.15 for both diesel and heating oil.

¹⁷ Early applications of this methodology are found in Ashenfelter and Card (1985); Card (1992); and Card and Krueger (1994, 2000); more recent applications in industrial economics include, for example, Ashenfelter, Hosken, and Weinberg (2013) and Genakos, Koutroumpis, and Pagliero (2018).

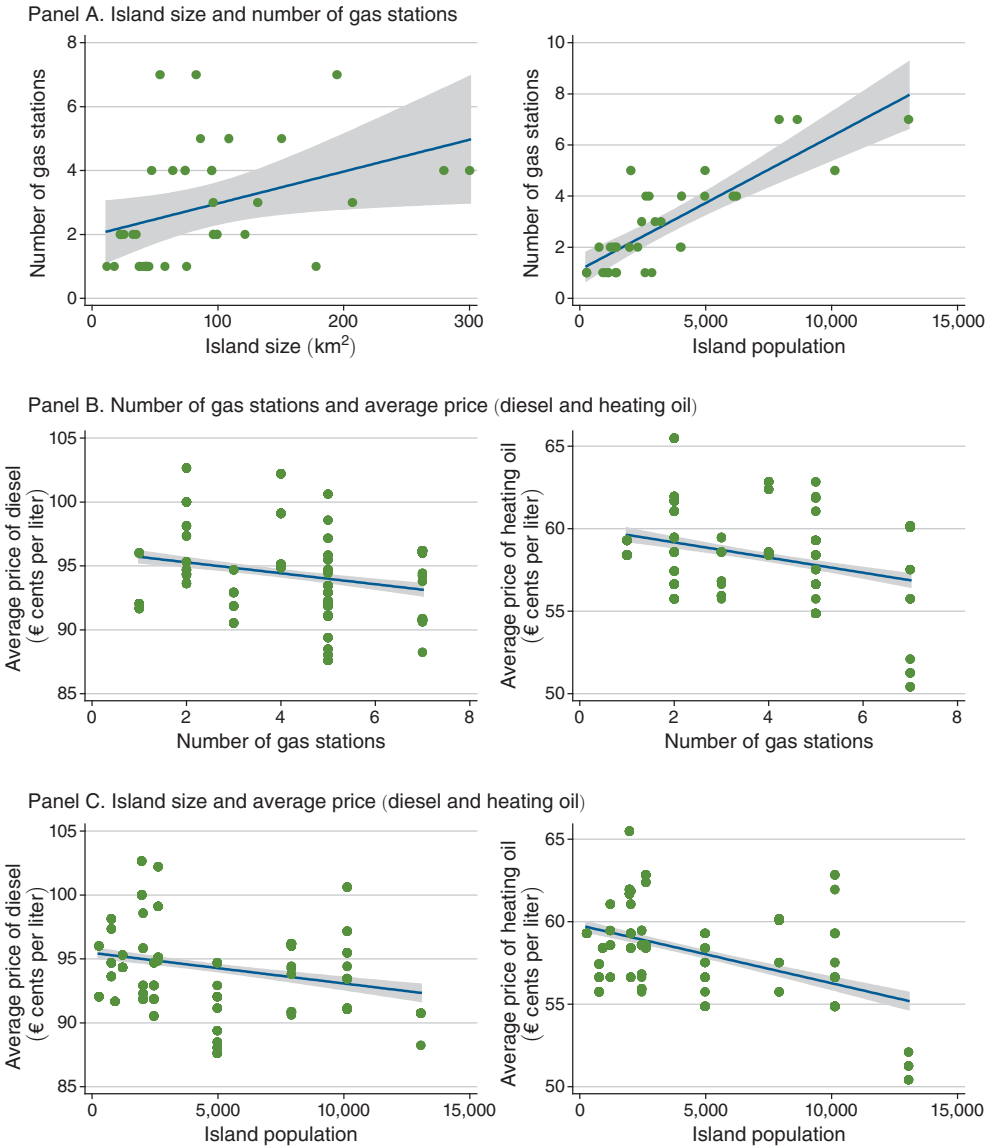


FIGURE 1. COMPETITION AND MARKET SIZE

Note: Average values are computed for January 2010 (before any excise duty change).

Source: Authors' calculations based on data from the Greek Ministry of Development and the Hellenic Statistical Authority.

framework. In fact, the tax increase was not anticipated and the prices of the different petroleum products tended to follow the same trend before the policy changes (Figure A1). In summary, the differential changes in excise duties across products (Table 1) provide identification of the tax pass-through, while fixed effects capture

island-specific and station-specific characteristics as well as the macroeconomic shocks that affected the whole economy and the control group accounts for aggregate changes in the prices of petroleum products.

Following Ashenfelter, Hosken, and Weinberg (2013), we conduct two tests of the parallel trend assumption. First, we estimate by OLS the equation

$$(4) \quad P_{kist} = \beta_0 + \gamma Trend_t + \gamma_T Trend_t \times Treat + \beta_k + \beta_s + e_{kist},$$

where P_{kist} denotes the retail price of product k , on island i , in gas station s , on day t , and $Treat$ is an indicator variable for products in the treatment group (diesel, gasoline, unleaded gasoline). We separately estimate (4) using data for the ten days before each excise duty change. We then test and cannot reject the null hypothesis that the coefficient γ_T is equal to zero at the 5 percent confidence level (Table A3). We also replace the treatment indicator with product-specific dummies and test the difference in trends between pairs of products. Again, we cannot reject the null hypothesis of parallel trends across products at the 5 percent confidence level (Table A3).

Second, we replace the trend variable in equation (4) with more flexible period-specific dummies β_t . We also replace the interaction of the trend and the treatment group indicator with $\beta_t \times Treat$ and then test the null hypothesis that the coefficients of the period-specific interactions are all equal to zero (individually and jointly). Even with this more flexible specification, we cannot reject the null hypothesis of parallel trends at the 5 percent confidence level (Table A4).¹⁸

Although variables in Z capture the potential effect of other observed island characteristics on pass-through, in Section IVB we will also report IV estimates of model (3), where exogenous variability in market size is used to estimate the impact of the number of competitors on pass-through. Following the literature on equilibrium entry in oligopoly markets (Bresnahan and Reiss 1991; Berry 1992; Mazzeo 2002; Toivanen and Waterson 2005), the rationale for the IV approach is that market size is a crucial determinant of entry and competition while it is arguably uncorrelated with unobservable determinants of the pass-through (such as demand convexity). Hence, the IV approach assumes that market size can be excluded from Z while being correlated with measures of competition. This second assumption can be tested, and it is verified in our results, described in the next section.

IV. Empirical Results

A. The Estimated Pass-Through

Figure 2 shows the average price difference between treated (diesel, unleaded 95, unleaded 100, and super) and control (heating oil) products for ten days before and after each change in excise duties. The solid lines represent linear regressions estimated separately before and after the tax changes. Figure A2 in the online Appendix

¹⁸ We also estimate these specifications using longer windows for tax changes 1 and 3. The results are not significantly affected.

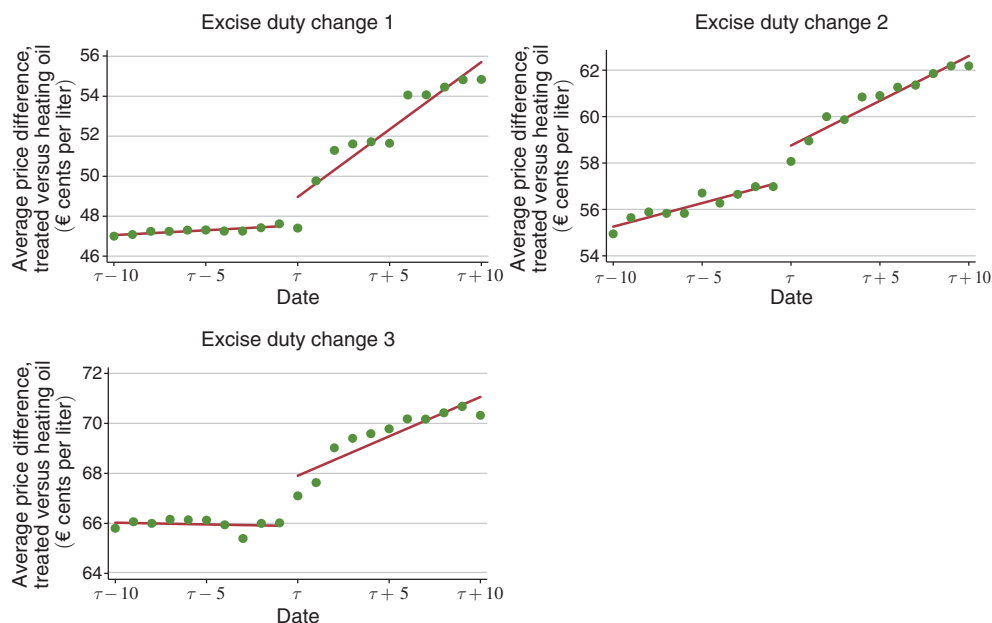


FIGURE 2. AVERAGE PRICE DIFFERENCES BETWEEN TREATED AND CONTROL PRODUCTS

Notes: The three figures plot the average price difference between treated (diesel, unleaded 95, unleaded 100, and super) and control (heating oil) products (for ten days before and after each change in excise duties) together with linear regression lines for the periods before and after the tax change. Figure A2 in the online Appendix plots similar graphs separately for each product and tax change.

Source: Authors' calculations based on data from the Greek Ministry of Development.

plots similar graphs separately for each product and tax change. There is a significant jump corresponding to the event date. Moreover, prices tend to increase during the days following the tax changes as stations progressively adjust their prices during this period.¹⁹ On average, 59 percent of product–station-specific prices are adjusted within three days, 88 percent within seven days, and 94 percent within ten days of the tax change.

The average pass-through on a given date depends on two margins. The extensive margin is the number of stations having adjusted their prices by a given date. The intensive margin is the size of the price increases for stations actually changing their prices. Accordingly, we can use equations (2) and (3) to estimate the average pass-through or the conditional pass-through (“conditional on starting to adjust”) using, respectively, all the data or only the data for firms that have changed their prices at least once by a given date. For long enough adjustment periods the two definitions coincide, as all stations have adjusted their prices. However, for shorter adjustment periods, the two definitions might differ substantially. We start by reporting results for the conditional and average pass-through for a ten-day adjustment

¹⁹ There are no significant changes in the price of heating oil around the changes in excise duties (see Figure A3 in the online Appendix).

TABLE 3—EXCISE DUTY PASS-THROUGH

Pass-through definition Dependent variable	Conditional $Price_{kist}$	Conditional $Price_{kist}$	Conditional $Price_{kist}$	Conditional $Price_{kist}$	Average $Price_{kist}$	Average $Price_{kist}$	Average $Price_{kist}$	Average $Price_{kist}$
Sample	Excise change 1 (1)	Excise change 2 (2)	Excise change 3 (3)	All excise changes (4)	Excise change 1 (5)	Excise change 2 (6)	Excise change 3 (7)	All excise changes (8)
Tax_{kt}	0.690 (0.131)	1.076 (0.101)	0.661 (0.106)	0.767 (0.099)	0.654 (0.129)	0.918 (0.122)	0.634 (0.112)	0.713 (0.103)
Observations	283	261	335	879	295	299	351	945
Within R^2	0.743	0.757	0.662	0.931	0.712	0.658	0.641	0.926
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Product $\times$ Station FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Excise change $\times$ Product type FE				Yes				Yes
Excise change $\times$ Station FE				Yes				Yes

Notes: The dependent variable is the retail price of product k , on island i , in gas station s , and on day $t \in \{\tau - 1, \tau + 10\}$, where τ is the date of each of the three excise duty changes. The pass-through is estimated using observations for station–product combinations that have changed the price at least once between τ and $\tau + 10$ in columns 1–4 (conditional pass-through) or all the available data in columns 5–8 (average pass-through). Standard errors clustered at the island level are reported in parentheses below coefficients.

Source: Authors' calculations based on data from the Greek Ministry of Development and Eurostat.

period, and in Section IVC we will compare the conditional and average pass-through for shorter adjustment periods. The ten-day adjustment period is chosen so that it is close enough to the change in the excise duty but also long enough for almost all of the gas stations to have changed their prices.²⁰

Table 3 reports the estimated coefficients of model (2). Columns 1–4 report the conditional pass-through, and columns 5–8 report the average pass-through for each tax change separately and pooling the data. The conditional pass-through is 0.77 (column 4) and the average pass-through is 0.71 (column 8), with standard errors of about 0.1.

Although excise duty changes vary significantly across products (Table 1), when we interact Tax_{kt} with product dummies, we cannot detect any significant difference in pass-through across products.²¹ The estimated pass-through is slightly smaller but within the range of the unit pass-through estimated by Marion and Muehlegger (2011); Chouinard and Perloff (2007); Doyle and Samphantharak (2008); and Alm, Sennoga, and Skidmore (2009) for US state taxes on petroleum products and by Poterba (1996) for sales taxes on clothing. This suggests that the market for petroleum products in our sample of islands does not operate very differently from other markets studied in the literature, which is important for the external validity of our results. The slightly lower value of pass-through in our case is likely related to the

²⁰ The likely cause of delayed price adjustments is the slow process of restocking gas stations in relatively remote areas. Restocking is done by ships that leave from Piraeus (the main port, near Athens) and follow a pre-determined route across the Aegean Sea. This process is determined by the geographic dispersion of islands in the Aegean Sea and is independent of the excise duty changes and the sizes of the islands (or other observable characteristics).

²¹ We also find that the pass-through is not significantly different for franchisees and independent gas stations.

TABLE 4—PASS-THROUGH AND COMPETITION

Estimation method	FE	FE	IV	FE	FE	IV
Dependent variable	$Price_{kist}$	$Price_{kist}$	$Price_{kist}$	$Price_{kist}$	$Price_{kist}$	$Price_{kist}$
Sample	All excise changes	All excise changes	All excise changes	All excise changes	All excise changes	All excise changes
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Conditional pass-through</i>						
Tax_{it}	0.449 (0.103)	−0.736 (1.095)	0.464 (0.109)	0.139 (0.208)	−0.601 (0.930)	−0.702 (0.535)
$Tax_{it} \times \text{Number of competitors}_s$	0.086 (0.024)	0.082 (0.033)	0.082 (0.023)	0.289 (0.112)	0.265 (0.135)	0.821 (0.348)
$Tax_{it} \times \text{Number of competitors}_s^2$				−0.025 (0.012)	−0.023 (0.015)	−0.090 (0.044)
First-stage F -test (number of competitors)			21.86			36.53
First-stage F -test (number of competitors ²)						13.21
Within R^2	0.937	0.939		0.939	0.940	
Observations	879	879	879	879	879	879
<i>Panel B. Average pass-through</i>						
Tax_{it}	0.409 (0.109)	−0.599 (1.107)	0.403 (0.129)	0.233 (0.228)	−0.561 (1.065)	−0.948 (0.865)
$Tax_{it} \times \text{Number of competitors}_s$	0.082 (0.023)	0.068 (0.036)	0.083 (0.024)	0.195 (0.132)	0.110 (0.158)	0.930 (0.566)
$Tax_{it} \times \text{Number of competitors}_s^2$				−0.014 (0.014)	−0.005 (0.018)	−0.104 (0.073)
First-stage F -test (number of competitors)			18.00			30.66
First-stage F -test (number of competitors ²)						11.34
Within R^2	0.931	0.934		0.931	0.934	
Observations	945	945	945	945	945	945
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Product $\times$ Station FE	Yes	Yes	Yes	Yes	Yes	Yes
Excise change $\times$ Product type FE	Yes	Yes	Yes	Yes	Yes	Yes
Excise change $\times$ Station FE	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls (interactions with income, education, number of ports and airports, distance from Piraeus, and tourist arrivals)		Yes			Yes	

Notes: The dependent variable is the retail price of product k , on island i , in gas station s , and on day $t \in \{\tau - 1, \tau + 10\}$, where τ is the date of each of the three excise duty changes. In panel A the pass-through is estimated using observations for station–product combinations that have changed the price at least once between τ and $\tau + 10$ (conditional pass-through), whereas in panel B we use all available data (average pass-through). Standard errors clustered at the island level are reported in parentheses below coefficients. First-stage results for columns 3 and 6 are reported in Table A9 of the online Appendix.

Source: Authors’ calculations based on data from the Greek Ministry of Development, the Hellenic Statistical Authority, and Eurostat.

fact that many gas stations in our data have a significant degree of market power. It is this topic that we explore next.

B. Pass-Through and Competition

Table 4, column 1 reports the results of model (3), allowing for the interaction between tax changes and number of competitors. In column 2, we add controls for the interaction of excise duty changes and island characteristics, such as income, education, number of ports and airports, distance from Piraeus, and number of tourist arrivals. The conditional and average pass-through increase significantly with competition. Column 4 shows that the relation between competition and pass-through is

concave. This result is robust, controlling for interactions of excise duty changes and covariates (column 5).²² Table 4, columns 3 and 6 report the IV estimates, where the instruments are island population and its square.²³ First-stage results (reported in Table A9 in the Appendix) are highly significant, showing a strong correlation between market size and number of competitors. Overall, the impact of competition on pass-through is positive and decreasing as the number of competitors grows.

The nonlinear relation between competition and pass-through is more clearly described in Figure 3, which shows the results of a nonparametric specification for the conditional pass-through (reported in Table 7, column 1). The pass-through is about 0.44 in monopoly islands and increases up to about 1 in islands with four competitors. The relation between pass-through and number of competitors is flat thereafter. Table A10 in the online Appendix reports the corresponding results for the average pass-through. The quick convergence to a unit pass-through is in line with the results of Bresnahan and Reiss (1991) that show that entry thresholds converge quite quickly; in other words, after three or four firms, an additional entrant does not affect competition much. Note that the estimated pass-through for monopoly islands is not significantly different from the 0.5 pass-through predicted by a monopoly model with linear demand.

C. Pass-Through and Speed of Adjustment

The results shown in Tables 3 and 4 are obtained with a ten-day adjustment period. Table 5 reports the estimated average and conditional pass-through for different adjustment periods. Shorter adjustment periods imply a lower average pass-through, as stations progressively adjust their prices. Figure 4 shows that the average pass-through converges to the conditional pass-through. The conditional pass-through does not depend significantly on the length of the adjustment period (Table 5, column 2). The speed of convergence of the average and the conditional pass-through is in line with the relatively fast exchange rate pass-through measured by Bonadio, Fischer, and Sauré (2020). Still, the speed of adjustment in our data is slower than that observed in other studies of the gasoline market (for example, Knittel, Meiselman, and Stock 2016). This can be partly explained by some specificities of our sample. In particular, the average pass-through in our sample can be affected by the lower frequency of restocking of gas stations on islands (relative to stations on the mainland).²⁴

Does the speed of adjustment depend on competition? This is an important question, as it relates to understanding how quickly prices adjust to cost shocks in the economy. In imperfectly competitive markets, we cannot expect an equal speed of adjustment in markets with different level of competition (Gopinath and Itskhoki 2010). We split the islands into two groups: those with one to three competitors (“low competition”) and those with four or more competitors (“high competition”).

²² Tables A5–A8 in the online Appendix provide results introducing interactions one by one.

²³ Results are robust using island size as the instrument.

²⁴ Using a probit model, we find that the probability of a price change is not systematically related to any island characteristic, such as size, population, or distance from Piraeus.

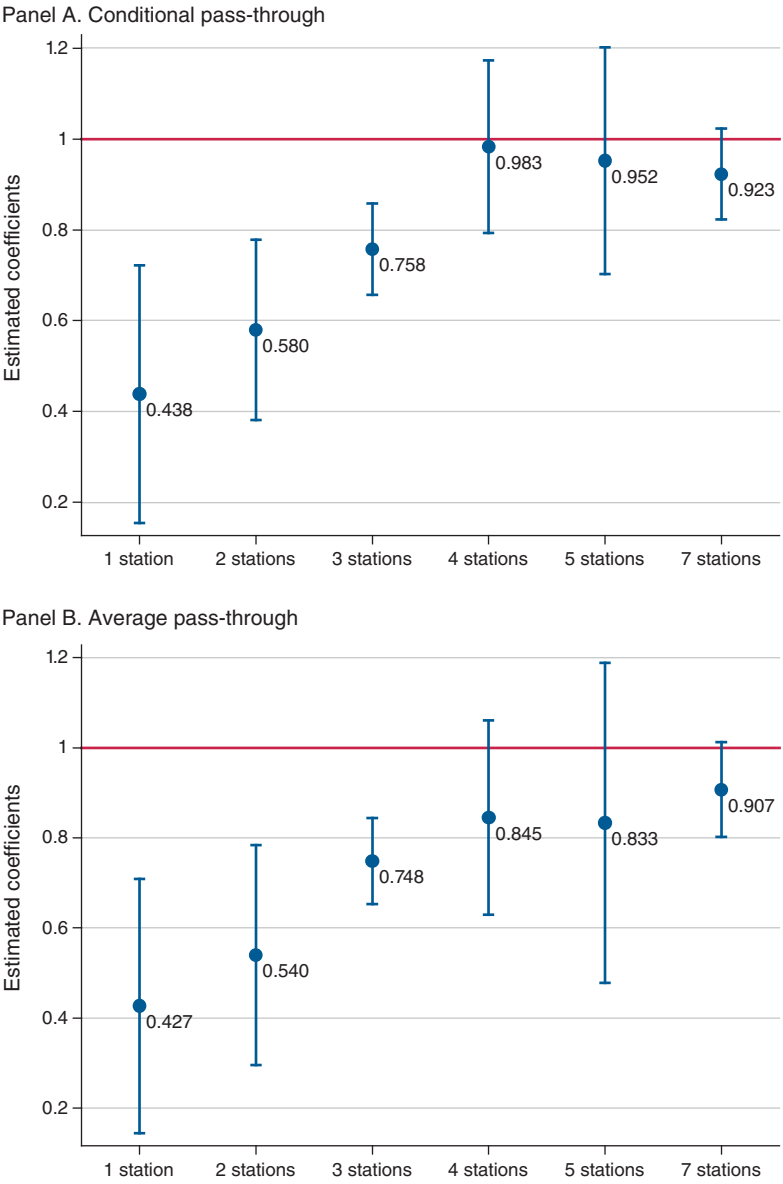


FIGURE 3. PASS-THROUGH AND COMPETITION

Notes: In panel A the figure plots the estimated coefficients from Table 7, column 1 together with the 95 percent confidence interval. In panel B the figure plots the estimated coefficients from Table A10, column 1 together with the 95 percent confidence interval.

Source: Authors' calculations based on data from the Greek Ministry of Development.

Table 6 and Figure 5 report the average and the conditional pass-through for the two groups for different adjustment periods. The average pass-through is significantly higher for islands with more competitors. At $t + 1$, the pass-through in more competitive markets is about 0.16 higher (about double) than in less competitive

TABLE 5—SPEED OF ADJUSTMENT

Dependent variable	$Price_{kist}$	$Price_{kist}$
Sample	All excise changes	All excise changes
Pass-through definition	Average	Conditional
	(1)	(2)
Tax_{it}	0.232	0.805
$(\tau - 1, \tau + 1)$	(0.111)	(0.134)
Tax_{it}	0.339	0.816
$(\tau - 1, \tau + 2)$	(0.101)	(0.129)
Tax_{it}	0.368	0.771
$(\tau - 1, \tau + 3)$	(0.099)	(0.120)
Tax_{it}	0.421	0.741
$(\tau - 1, \tau + 4)$	(0.091)	(0.109)
Tax_{it}	0.417	0.727
$(\tau - 1, \tau + 5)$	(0.091)	(0.109)
Tax_{it}	0.596	0.732
$(\tau - 1, \tau + 6)$	(0.114)	(0.111)
Tax_{it}	0.618	0.687
$(\tau - 1, \tau + 7)$	(0.118)	(0.110)
Tax_{it}	0.667	0.727
$(\tau - 1, \tau + 8)$	(0.117)	(0.113)
Tax_{it}	0.707	0.759
$(\tau - 1, \tau + 9)$	(0.105)	(0.102)
Tax_{it}	0.713	0.767
$(\tau - 1, \tau + 10)$	(0.103)	(0.099)

Notes: Each coefficient comes from a separate regression. The dependent variable is the retail price of product k , on island i , in gas station s , and on day $t \in \{\tau - 1, \tau + \delta\}$, where τ is the date of each of the three excise duty changes and $\delta = 1, \dots, 10$ is the adjustment period. The fixed effects are the same as in Table 3, columns 4 and 8. The average pass-through (column 1) is estimated using all the data. The conditional pass-through (column 2) is estimated using observations for station–product combinations that have changed the price at least once between τ and $\tau + \delta$. Standard errors clustered at the island level are reported in parentheses below coefficients.

Source: Authors’ calculations based on data from the Greek Ministry of Development and Eurostat.

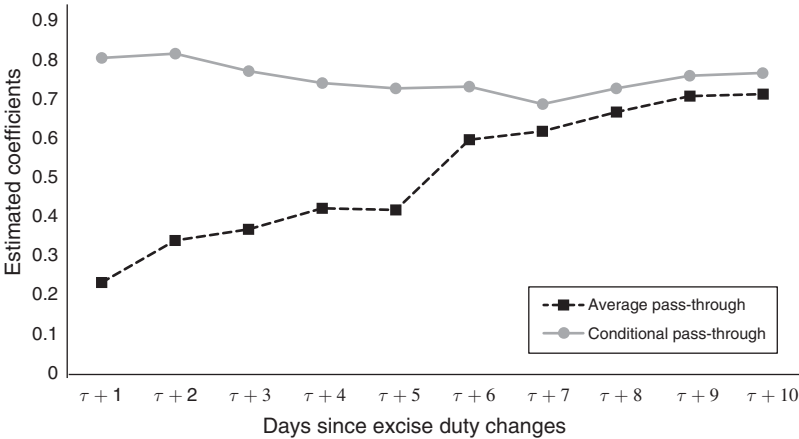


FIGURE 4. PASS-THROUGH AND SPEED OF ADJUSTMENT

Notes: The figure plots the estimated coefficients from Table 5. The average pass-through is estimated using all the data. The conditional pass-through is estimated using observations for station–product combinations that have changed the price at least once between τ and $\tau + \delta$, where τ is the date of the excise duty change and $\delta = 1, \dots, 10$.

Source: Authors’ calculations based on data from the Greek Ministry of Development.

TABLE 6—SPEED OF ADJUSTMENT AND COMPETITION

<i>Panel A. Average pass-through</i>					
Dependent variable	$Price_{kist}$	$Price_{kist}$	$Price_{kist}$	$Price_{kist}$	$Price_{kist}$
Sample	All excise changes	All excise changes	All excise changes	All excise changes	All excise changes
	$(\tau - 1, \tau + 1)$	$(\tau - 1, \tau + 2)$	$(\tau - 1, \tau + 3)$	$(\tau - 1, \tau + 4)$	$(\tau - 1, \tau + 5)$
	(1)	(2)	(3)	(4)	(5)
$Tax_{it} \times$ Low competition (1–3 competitors)	0.136 (0.074)	0.200 (0.082)	0.198 (0.078)	0.273 (0.065)	0.268 (0.065)
$Tax_{it} \times$ High competition (4–7 competitors)	0.301 (0.152)	0.433 (0.129)	0.500 (0.118)	0.534 (0.111)	0.534 (0.111)
Test equality of coefficients (p -value)	0.230	0.060	0.009	0.011	0.010
Dependent variable	$Price_{kist}$	$Price_{kist}$	$Price_{kist}$	$Price_{kist}$	$Price_{kist}$
Sample	All excise changes	All excise changes	All excise changes	All excise changes	All excise changes
	$(\tau - 1, \tau + 6)$	$(\tau - 1, \tau + 7)$	$(\tau - 1, \tau + 8)$	$(\tau - 1, \tau + 9)$	$(\tau - 1, \tau + 10)$
	(6)	(7)	(8)	(9)	(10)
$Tax_{it} \times$ Low competition (1–3 competitors)	0.410 (0.105)	0.443 (0.104)	0.456 (0.103)	0.519 (0.094)	0.531 (0.092)
$Tax_{it} \times$ High competition (4–7 competitors)	0.747 (0.125)	0.766 (0.135)	0.831 (0.118)	0.855 (0.107)	0.856 (0.107)
Test equality of coefficients (p -value)	0.014	0.021	0.006	0.008	0.011
<i>Panel B. Conditional pass-through</i>					
Dependent variable	$Price_{ist}$	$Price_{ist}$	$Price_{ist}$	$Price_{ist}$	$Price_{ist}$
Sample	All excise changes	All excise changes	All excise changes	All excise changes	All excise changes
	$(\tau - 1, \tau + 1)$	$(\tau - 1, \tau + 2)$	$(\tau - 1, \tau + 3)$	$(\tau - 1, \tau + 4)$	$(\tau - 1, \tau + 5)$
	(1)	(2)	(3)	(4)	(5)
$Tax_{it} \times$ Low competition (1–3 competitors)	0.639 (0.110)	0.614 (0.092)	0.528 (0.078)	0.528 (0.072)	0.523 (0.072)
$Tax_{it} \times$ High competition (4–7 competitors)	0.888 (0.120)	0.952 (0.082)	0.966 (0.058)	0.953 (0.060)	0.932 (0.067)
Test equality of coefficients (p -value)	0.063	0.000	0.000	0.000	0.000
Dependent variable	$Price_{ist}$	$Price_{ist}$	$Price_{ist}$	$Price_{ist}$	$Price_{ist}$
Sample	All excise changes	All excise changes	All excise changes	All excise changes	All excise changes
	$(\tau - 1, \tau + 6)$	$(\tau - 1, \tau + 7)$	$(\tau - 1, \tau + 8)$	$(\tau - 1, \tau + 9)$	$(\tau - 1, \tau + 10)$
	(6)	(7)	(8)	(9)	(10)
$Tax_{it} \times$ Low competition (1–3 competitors)	0.509 (0.105)	0.486 (0.096)	0.502 (0.093)	0.552 (0.085)	0.565 (0.082)
$Tax_{it} \times$ High competition (4–7 competitors)	0.939 (0.091)	0.886 (0.095)	0.926 (0.089)	0.948 (0.076)	0.951 (0.076)
Test equality of coefficients (p -value)	0.001	0.001	0.001	0.000	0.001

Notes: The dependent variable is the retail price of product k , on island i , in gas station s , and on day $t \in \{\tau - 1, \tau + \delta\}$, where τ is the date of each of the three excise duty changes and $\delta = 1, \dots, 10$. The average pass-through is estimated using all the data. The conditional pass-through is estimated using observations for station-product combinations that have changed the price at least once between τ and $\tau + \delta$. Standard errors clustered at the island level are reported in parentheses below coefficients. The p -value of the test of equality of each set of coefficients is reported in the last row of each panel.

Source: Authors' calculations based on data from the Greek Ministry of Development and Eurostat.

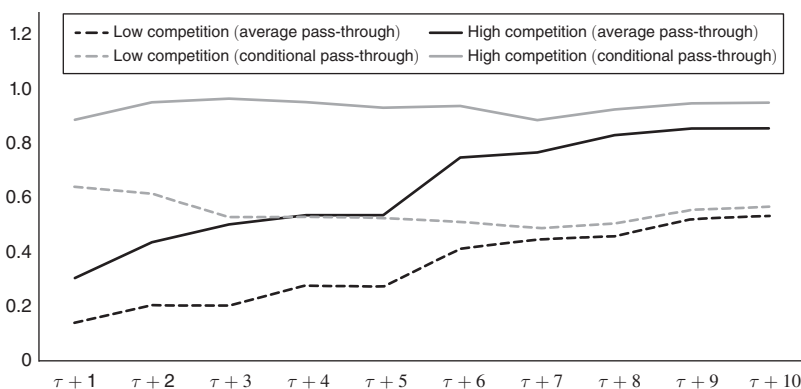


FIGURE 5. SPEED OF ADJUSTMENT AND COMPETITION

Notes: The figure plots the average and conditional pass-through on islands with one to three (“low competition”) and four to seven (“high competition”) gas stations. The average pass-through is estimated using all the data. The conditional pass-through is estimated using observations for station–product combinations that have changed the price at least once between τ and $\tau + \delta$, where τ is the date of the excise duty change and $\delta = 1, \dots, 10$. Estimated coefficients are reported in Table 6.

Source: Authors’ calculations based on data from the Greek Ministry of Development.

markets. At $t + 10$, the pass-through in more competitive markets is about 0.3 (or 60 percent) higher. The conditional pass-through is stable over time and significantly larger in more competitive markets. Finally, Figure 6 shows the cumulative frequency of price changes for the two groups and provides direct evidence that stations in more competitive markets react more quickly to changes in excise duties.²⁵ Hence, more competitive markets adjust faster to cost shocks, partly because the conditional pass-through is larger and partly because price changes are faster.

These results imply a positive correlation between pass-through and the frequency of price adjustments across islands (with different levels of competition). This is consistent with the results of Gopinath and Itskhoki (2010), who find a positive correlation between the frequency of price adjustments and the magnitude of exchange-rate pass-through across sectors. Our results are also in line with their theoretical model, in which firms in more competitive markets (i.e., with higher residual demand elasticity) are those with a higher frequency of price adjustments and higher pass-through.

One might argue that the higher frequency of price adjustments in more competitive islands might be driven by the higher frequency at which these stations are restocked. While we do not have direct evidence on the frequency of restocking, we can study two instances of changes in marginal costs that cannot possibly be affected by the frequency of restocking. In particular, we collected the same data as we used in the main analysis for two changes in VAT (from 19 percent to 21 percent

²⁵ The Kolmogorov-Smirnov test rejects equality of the CDFs at the 1 percent confidence level for all the excise incidents (cumulatively) and for each one separately.

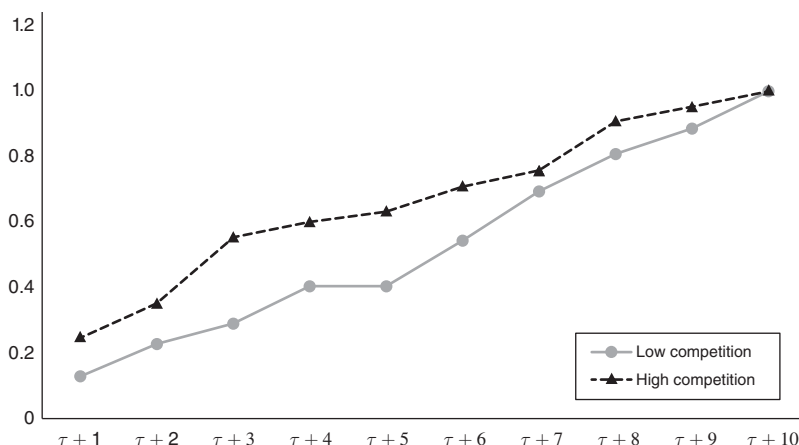


FIGURE 6. CUMULATIVE FREQUENCY OF PRICE CHANGES

Notes: The figure plots the cumulative frequency of station–product combinations that changed their prices between τ and $\tau + \delta$, where τ is the date of the excise duty change and $\delta = 1, \dots, 10$, on islands with one to three (“low competition”) and four to seven (“high competition”) gas stations. The Kolmogorov-Smirnov test rejects the equality of the CDFs at the 1 percent confidence level.

on March 15, 2010 and from 21 percent to 23 percent on July 1, 2010).²⁶ We compute the frequency of price changes after the VAT changes and plot the cumulative distribution for low-competition and high-competition islands in Figure A4 of the online Appendix. If the results in Figure 6 were driven by the frequency at which islands were restocked, we would observe no difference in the empirical CDFs for VAT changes. However, Figure A4 provides the same qualitative picture, suggesting that the frequency of restocking is not the driving force behind our previous results.²⁷

D. Alternative Market Definitions

Without a clear definition of market boundaries or detailed traffic data (Houde 2012), the literature has typically defined markets based on the distance between gas stations (Shepard 1991; Barron, Taylor, and Umbeck 2004; Eckert and West 2005; Hosken, McMillan, and Taylor 2008; Pennerstorfer et al. 2020). Competition authorities throughout the world routinely define markets based on geographic or driving distance between sellers.²⁸ While realistic, this approach cannot guarantee the absence of substitution effects for firms outside the geographic area considered. In our application, the definition of markets is simpler. Monopoly islands are

²⁶ VAT contributes to the definition of retail prices as described in Section II. VAT changes affect all products in the same way, so these events do not provide us with a control group. Hence, we do not use these two changes in the main analysis.

²⁷ The Kolmogorov-Smirnov test rejects equality of the CDFs at the 1 percent confidence level for all the VAT changes (cumulatively) and for each one separately.

²⁸ See, for example, EU Commission Case M.7603 – Statoil Fuel and Retail/Dansk Fuels and UK Competition and Markets Authority decision ME/6534/15.

unambiguously classified as such. In islands with more stations, there can only be substitution effects among firms on the same island. Given the small scale of these islands, some substitution is likely to exist among all stations on the same island.

We followed standard market definitions and computed for each station the number of competitors within a three-kilometer radius, a three-kilometer and a five-kilometer driving distance, and a ten-minute drive (using Google Maps).²⁹ While these three procedures obviously do not affect monopoly islands, they may reduce the number of competitors for stations on larger islands. We then estimate the pass-through using model (3) for stations that have a different number of competitors according to the new market definitions. Table 7 reports the estimated coefficients for the conditional pass-through, and Table A10 in the online Appendix reports the corresponding coefficients for the average pass-through.

The comparison of the estimated pass through in monopoly islands (the first row in Table 7) is particularly telling. In this case, we know that the island market definition (column 1) correctly classifies monopoly islands, which implies that the estimated pass-through is the correct benchmark for evaluating the other market definitions. The fact that the pass-through in columns 2–5 is significantly higher indicates that the alternative distance definitions incorrectly classify some markets as monopolies, ignoring substitution effects across stations and leading to upward-biased coefficients that are 70 percent, 59 percent, 60 percent, or 57 percent higher. This shows that the distance definition leads to a substantial bias in the estimated pass-through.

Table 7 shows that similar results hold for duopolies and triopolies. The pass-through is systematically higher using the distance definitions. For example, the three-kilometer driving distance definition implies that the pass-through is 64 percent and 36 percent higher for markets with two and three firms, respectively. This difference becomes negligible in markets with four or more sellers. Given the results obtained for monopoly islands, the most likely explanation for this difference is the existence of substitution effects across sellers on the same island, which is not captured by the distance definitions.

However, we cannot precisely quantify the bias induced by the distance definition for duopolies and triopolies. This is because it is not possible to identify the correct benchmark in this case. For example, the island market definition might incorrectly classify as duopolies some islands on which there are in fact two monopolies (two stations that are sufficiently far not to interact with each other). On the other hand, the distance definition might incorrectly classify as duopolies pairs of gas stations on islands with three or more firms, which are actually competing with each other. Hence, neither of these definitions is necessarily correct in this case. Although the distance definition implies an overestimation of the pass-through, we cannot interpret the difference in estimated coefficients as a measure of the bias.

Overall, our results suggest that care should be taken when using the standard approach based on distance between sellers, particularly when there is a small number of competitors. This is relevant for academic research and competition policy alike.

²⁹ A continuum of possible market definitions exists, but we restricted attention to definitions often used in practice.

TABLE 7—CONDITIONAL PASS-THROUGH AND COMPETITION: ALTERNATIVE MARKET DEFINITIONS

Dependent variable	$Price_{kist}$	$Price_{kist}$	$Price_{kist}$	$Price_{kist}$	$Price_{kist}$
Market definition	Island (1)	3-km driving distance (2)	3-km radius (3)	10-min. driving distance (4)	5-km driving distance (5)
$Tax_{it} \times \text{One competitor}$	0.438 (0.137)	0.748 (0.105)	0.695 (0.084)	0.701 (0.093)	0.688 (0.085)
$Tax_{it} \times \text{Two competitors}$	0.580 (0.096)	0.951 (0.060)	1.046 (0.053)	0.915 (0.089)	0.972 (0.093)
$Tax_{it} \times \text{Three competitors}$	0.758 (0.048)	1.034 (0.115)	0.968 (0.089)	0.875 (0.067)	0.890 (0.060)
$Tax_{it} \times \text{Four competitors}$	0.983 (0.092)	1.020 (0.118)	1.034 (0.098)	0.963 (0.146)	1.009 (0.115)
$Tax_{it} \times \text{Five competitors}$	0.952 (0.120)	0.829 (0.059)	0.895 (0.072)	0.916 (0.065)	0.922 (0.061)
$Tax_{it} \times \text{Six competitors}$				0.794 (0.065)	0.814 (0.059)
$Tax_{it} \times \text{Seven competitors}$	0.923 (0.048)				
Observations	879	609	609	499	537
Within R^2	0.939	0.966	0.967	0.964	0.964
Time FE	Yes	Yes	Yes	Yes	Yes
Product $\times$ Station FE	Yes	Yes	Yes	Yes	Yes
Excise incident $\times$ Product type FE	Yes	Yes	Yes	Yes	Yes
Excise incident $\times$ Station FE	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is the retail price of product k , on island i , in gas station s , and on day $t \in \{\tau - 1, \tau + \delta\}$, where τ is the date of each of the three excise duty changes. The pass-through is estimated using observations for station–product combinations that have changed the price at least once between τ and $\tau + 10$ (conditional pass-through). Standard errors clustered at the island level are reported in parentheses below coefficients. Table A10 in the online Appendix reports the results for the average pass-through.

Source: Authors' calculations based on data from the Greek Ministry of Development, the Hellenic Statistical Authority, Eurostat, and Google Maps.

V. Concluding Remarks

This paper provides new empirical evidence on the effects of competition on pass-through in clearly defined oligopolistic markets with a small number of firms. We contribute to the growing literature on pass-through by showing that pass-through increases with competition in a nonlinear fashion, growing from 0.4 for monopoly markets to about 1 for markets with four or more competitors. Moreover, the frequency of price adjustments is higher in more competitive markets. This might have important implications for the pass through at the macro level. We also find that conventional definitions of markets that are based on distance between sellers lead to overestimation of the pass-through for markets with up to four competitors. Since these definitions are often used in policy analysis, care should be taken when studying oligopolistic markets.

We acknowledge that the markets for petroleum products on Greek islands are not necessarily representative of oligopolistic markets for other products and on the mainland. We selected this environment precisely because it provides clean variation in competitive environments, which is not typically available for other markets.

The results contribute to our understanding of pass-through by showing new evidence on relationships that may be present in other settings and in larger markets.

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